

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for questions 1-10

1 Consider the statement:

If an animal can fly, then it is a bird.

Which of the following is the contrapositive of this statement?

- A. If an animal is not a bird, then it cannot fly.
- B. If an animal is a bird, then it can fly.
- C. If an animal can fly, then it is not a bird.
- D. If an animal cannot fly, then it is not a bird.

2 What is the equation of the line which passes through $A(4, 6, 5)$ and $B(-2, 3, -1)$?

A. $\tilde{r} = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} - \lambda \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix}$

B. $\tilde{r} = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} - \lambda \begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix}$

C. $\tilde{r} = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 9 \\ 4 \end{pmatrix}$

D. $\tilde{r} = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -3 \\ -6 \end{pmatrix}$

3 Consider the following proposition:

If a positive integer is not a perfect square, then it is not prime.

A counter example to this proposition is to be found. A number with which of the following properties would suffice?

- A. A positive integer that is not a perfect square and is prime.
- B. A positive integer that is not a perfect square and is not prime.
- C. A positive integer that is a perfect square and is prime.
- D. A positive integer that is a perfect square and is not prime.

4 A particle is moving in SHM in a straight line with an amplitude of 3 metres.

When the particle is 2 metres from the centre of motion, its speed is $5m/s$.

What is the period of the motion?

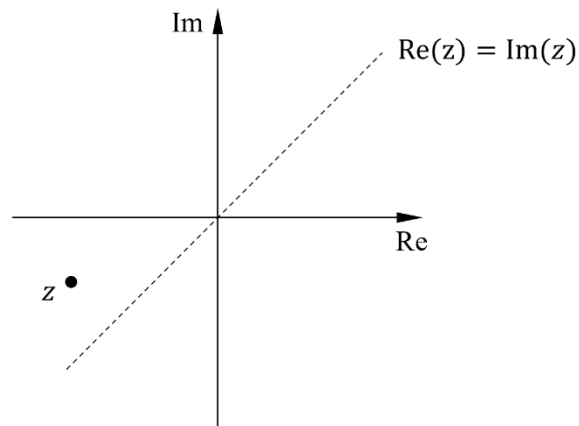
- A. $\frac{\sqrt{5}\pi}{2}$
- B. $\frac{2\sqrt{5}\pi}{5}$
- C. $\sqrt{5}\pi$
- D. $2\sqrt{5}\pi$

5 The roots of the cubic equation $f(x) = 0$ are α, β and γ .

Which of the following cubic equations has roots $2\alpha + 1, 2\beta + 1$ and $2\gamma + 1$?

- A. $f(2x + 1) = 0$
- B. $f\left(\frac{x}{2} + 1\right) = 0$
- C. $f\left(\frac{x-1}{2}\right) = 0$
- D. $f\left(\frac{x+1}{2}\right) = 0$

- 6 The diagram below shows the complex number $z = x + iy$, which lies in the third quadrant such that $x < y$.



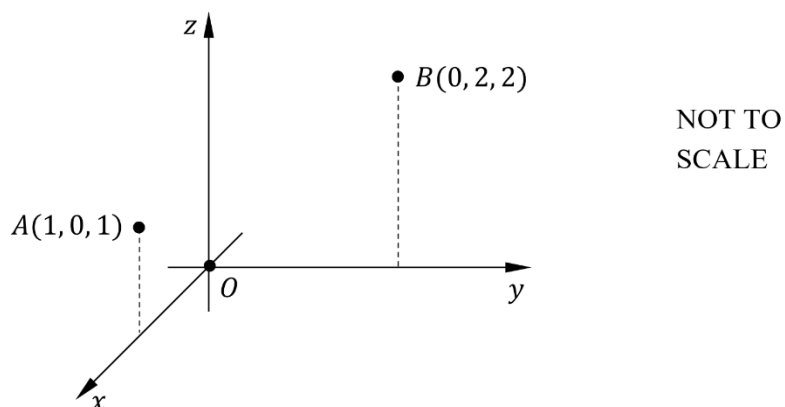
Given that $z^2 = a + ib$, which of the following statements is true?

- A. $a < 0, b < 0$
 - B. $a < 0, b > 0$
 - C. $a > 0, b < 0$
 - D. $a > 0, b > 0$
- 7 Consider the position vector of a particle $\vec{r} = t^2 \vec{i} + 2\cos t \vec{j} + 2\sin t \vec{k}$.

Which of the following best describes the motion of the particle?

- A. Spiralling at an increasing rate about the z axis in an anticlockwise direction.
- B. Spiralling at an increasing rate about the x axis in an anticlockwise direction.
- C. Spiralling at an increasing rate about the x axis in a clockwise direction.
- D. Spiralling at a constant rate about the x axis in an anticlockwise direction.

- 8 Consider triangle AOB , where A is the point $(1, 0, 1)$ and B is the point $(0, 2, 2)$.



Which of the following is true?

- A. $\angle AOB$ is a right angle
 - B. $\triangle AOB$ is isosceles
 - C. $\angle OAB$ is twice the size of $\angle OBA$
 - D. $\angle AOB$ is twice the size of $\angle OBA$
- 9 Consider the complex equation $z^3 = -p^3i$, where p is a positive, real number.

What is the sum of all of the imaginary parts of all of the solutions that have a negative principal argument?

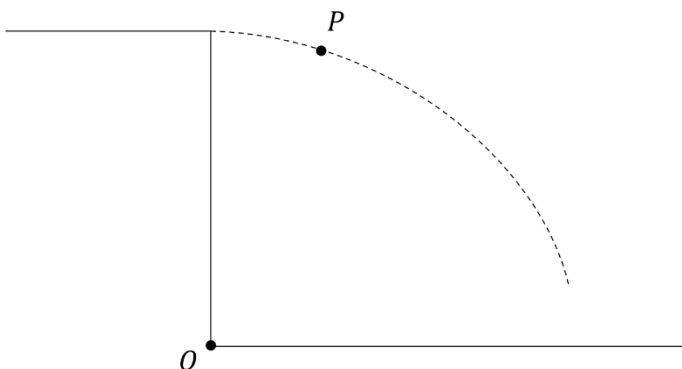
- A. pi
- B. $-pi$
- C. p
- D. $-p$

- 10 A unit mass is projected horizontally from the top of a tall building. Air resistance is R and the gravitational force is G .

The position vector of the mass is $\mathbf{r}$, with the origin being the base of the building.

The velocity vector of the mass is $\mathbf{v}$.

At the point P , $\mathbf{r} \cdot \mathbf{v} = 0$.



Which of the following is consistent with the motion of the particle at P ?

- A. $\frac{\mathbf{R} \cdot \mathbf{v}}{|\mathbf{R}||\mathbf{v}|} < \frac{\mathbf{G} \cdot \mathbf{r}}{|\mathbf{G}||\mathbf{r}|} < \frac{\mathbf{R} \cdot \mathbf{r}}{|\mathbf{R}||\mathbf{r}|} < \frac{\mathbf{G} \cdot \mathbf{v}}{|\mathbf{G}||\mathbf{v}|}$
- B. $\frac{\mathbf{G} \cdot \mathbf{r}}{|\mathbf{G}||\mathbf{r}|} < \frac{\mathbf{G} \cdot \mathbf{v}}{|\mathbf{G}||\mathbf{v}|} < \frac{\mathbf{v} \cdot \mathbf{R}}{|\mathbf{v}||\mathbf{R}|} < \frac{\mathbf{R} \cdot \mathbf{r}}{|\mathbf{R}||\mathbf{r}|}$
- C. $\frac{\mathbf{r} \cdot \mathbf{R}}{|\mathbf{r}||\mathbf{R}|} < \frac{\mathbf{v} \cdot \mathbf{G}}{|\mathbf{v}||\mathbf{G}|} < \frac{\mathbf{r} \cdot \mathbf{G}}{|\mathbf{r}||\mathbf{G}|} < \frac{\mathbf{v} \cdot \mathbf{R}}{|\mathbf{v}||\mathbf{R}|}$
- D. $\frac{\mathbf{v} \cdot \mathbf{G}}{|\mathbf{v}||\mathbf{G}|} < \frac{\mathbf{r} \cdot \mathbf{G}}{|\mathbf{r}||\mathbf{G}|} < \frac{\mathbf{R} \cdot \mathbf{v}}{|\mathbf{R}||\mathbf{v}|} < \frac{\mathbf{r} \cdot \mathbf{R}}{|\mathbf{r}||\mathbf{R}|}$

End of Section I

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) (Start a new writing booklet.)

Marks

- a. Consider the vector $\underline{r} = p\underline{i} - 4\underline{j} + 5\underline{k}$ 2

If $\underline{r}$ has a magnitude of $5\sqrt{2}$ units, what is the exact value/s of p ?

- b. Find the size of the angle between the vectors 2

$$\underline{a} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \text{ and } \underline{b} = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$$

giving your answer correct to the nearest minute.

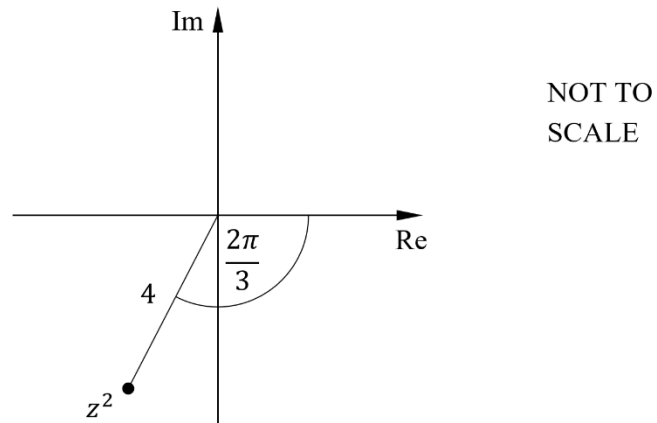
- c. Sketch the region in the complex plane where the inequalities 3
 $|z - i| \leq 1$ and $0 \leq \arg(z - i) \leq \frac{\pi}{3}$ hold simultaneously.

- d. Show that, if a and b are positive integers, then $\frac{a}{b} + \frac{b}{a} \geq 2$ 2

Question 11 continued on next page.

Question 11 continued.

- e. The complex number z^2 is shown on the diagram below. 2



Given that z lies in the fourth quadrant, find z in mod-arg form.

- f. Consider the equation $(p + 5)^{36} + (p + 3)^{35} = 7$. 2

Give reasons as to why there is no integer p root for the equation.

- g. Find: 2

$$\int x \ln x \, dx$$

End of Question 11

Question 12 (15 marks) (Start a new writing booklet.)**Marks**

- a. i) Find A , B , and C such that: 2

$$\frac{x}{(x-2)^2(x-1)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x-1}.$$

- ii) Hence, show that: 3

$$\int_3^4 \frac{x}{(x-2)^2(x-1)} dx = \ln\left(\frac{3}{4}\right) + 1.$$

- b. Consider the statement P : 2

“If $2n^2 + 3n + 1$ is even, then n is odd.”

Prove that the converse of P is true.

- c. Factorise the equation $P(z) = z^4 - 6z^3 + 18z^2 - 14z - 39$, given that 3
one of the zeros of $P(z)$ is $(2 - 3i)$.

- d. Consider the complex number $z = re^{i\theta}$: 2

Show that $\phi = \frac{2z}{z^2 + |z|^2}$ is purely real.

- e. Find: 3

$$\int \frac{e^{2x} - e^x}{e^x + 1} dx$$

End of Question 12

Question 13 (15 marks) (Start a new writing booklet.)**Marks**

- a. For the positive real numbers a and b , $a \neq b$, it is given that

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) \text{ and } (a - b)^2 > 0.$$

- i) Using these facts, or otherwise, prove that:

1

$$a^3 + b^3 > ab(a + b)$$

- ii) Hence, or otherwise, show that if c is also a positive real number

2

that is not equal to a or b , then:

$$2(a^3 + b^3 + c^3) > a^2(b + c) + b^2(a + c) + c^2(a + b)$$

- b. A particle moves along the x axis in a simple harmonic motion.

Its displacement, x cm, from the origin at time t seconds is given by:

$$x = 5 - 4\cos^2 2\pi t$$

- i) Find the centre and the amplitude of the motion.

3

- ii) Find the distance travelled by the particle in the first 5 seconds.

2

Question 13 continued on next page.

Question 13 continued.

- c. Prove by contradiction that 3

$$P(x) = 120 + 120x + 60x^2 + 20x^3 + 5x^4 + x^5$$

has no double roots.

- d. Let $z = 1 + i$ and $w = -\sqrt{3} + i$

- i) Find zw in modulus argument form. 2

- ii) Hence, show that $\cos\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}+\sqrt{2}}{4}$ 2

End of Question 13

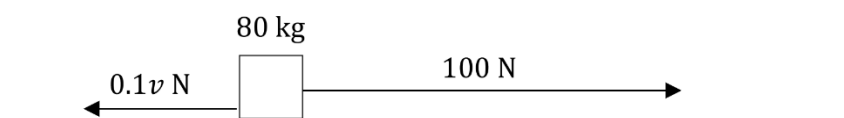
Question 14 (15 marks) (Start a new writing booklet.)**Marks**

a. Find:

3

$$\int \frac{2x-5}{x^2+2x+2} dx$$

b. An 80-kilogram mass is being dragged along a horizontal surface by a force of 100 newtons.



The mass is subject to friction of $0.1v$ newtons, where v is the speed of the mass in metres per second. The mass is initially at rest.

The equation of motion is

$$\ddot{x} = \frac{1000 - v}{800}. \quad (\text{DO NOT PROVE THIS.})$$

i) Show that the equation for time as a function of velocity is

3

$$t = 800 \ln \left(\frac{1000}{1000 - v} \right).$$

ii) Find the velocity of the mass after 10 seconds, correct to 1 decimal place.

3

Question 14 continued on next page.

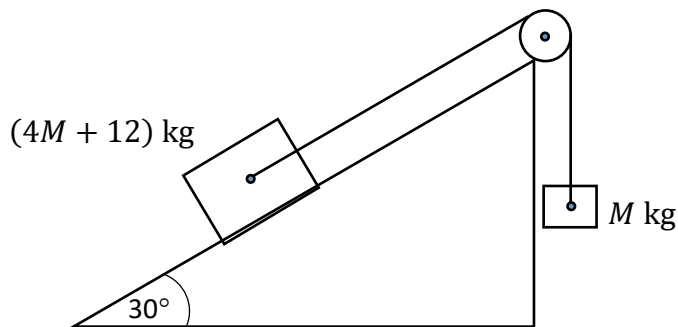
Question 14 continued.

- c. The vectors $\vec{u}$ and $\vec{v}$ are defined as $\vec{u} = \vec{a} + m\vec{b}$ and $\vec{v} = m\vec{a} - \vec{b}$, 3
where $\vec{a}$ and $\vec{b}$ are unit vectors and m is a real constant.

Given that the dot product of $\vec{u}$ and $\vec{v}$ is 1, and that the acute angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, find the value/s of m .

- d. Two masses M kg and $(4M + 12)$ kg are attached to ends of a light 3
inextensible string passing through a pulley.

The mass M kg is accelerating upwards at 3 ms^{-2} while the $(4M + 12)$ kg mass is moving downwards on a smooth inclined plane at angle 30° as shown.



The acceleration due to gravity is 10 ms^{-2} .

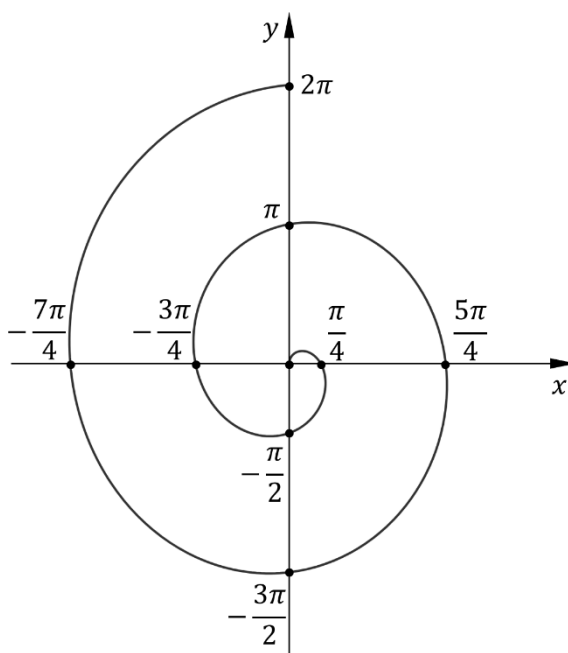
What is the size of the mass M ?

End of Question 14

Question 15 (15 marks) (Start a new writing booklet.)**Marks**

- a. A spiral starts at the origin and completes two clockwise spirals to end at $(0, 2\pi)$.

Its position from $t = 0$ to $t = T$ is shown in the diagram.



The parametric equations of the spiral are:

$$\begin{cases} x = f(t) \\ y = g(t) \end{cases} \quad \text{for } [0, T].$$

- i) Find $f(t)$, $g(t)$ and T .

3

- ii) The parametric equations of a second spiral are:

1

$$\begin{cases} x = g(t) \\ y = f(t) \end{cases} \quad \text{for } \left[0, \frac{T}{2}\right].$$

If the second spiral was drawn on the axes above, how many points of intersection would it have with the first spiral?

Question 15 continued on next page.

Question 15 continued.

- b. Consider the integral

$$I_n = \int_0^1 \sqrt{x}(1-x)^n dx, \quad n = 0, 1, 2, 3, \dots$$

- i) Show that

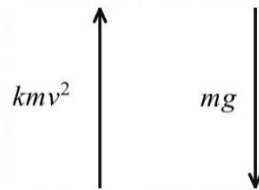
3

$$I_n = \frac{2n}{2n+3} I_{n-1}, \quad n = 1, 2, 3, \dots$$

- ii) Evaluate I_1 .

1

- c. A particle of mass m is released and allowed to fall vertically in a medium in which the resistance is mk times the square of the velocity (v) of the particle.



- i) Write an equation for the acceleration of the particle in terms of v .

1

- ii) Show that the terminal velocity (V_T) is given by $V_T = \sqrt{\frac{g}{k}}$.

1

- iii) Show that the distance fallen (x) in terms of v is:

3

$$x = \frac{1}{2k} \ln \left(\frac{g}{g - kv^2} \right)$$

- iv) Hence, find an expression for the distance that the particle has fallen

2

when the velocity reaches $\frac{1}{2}V_T$.

End of Question 15

Question 16 (15 marks) (Start a new writing booklet.)**Marks**

- a. The line L with equation $\vec{r} = \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ intersects with the sphere 4
with equation $(x + 1)^2 + (y - 2)^2 + (z - 3)^2 = 45$ at the two points,
 A and B .

Find the shortest distance between the centre of the sphere C and
the chord AB .

- b. Given $a_{n+1} = \sqrt{2a_n}$ and $a_1 = 1$, prove by mathematical induction that 3
for all integers $n \geq 1$

$$a_n = 2^{(1-2^{(1-n)})}$$

- c. i) Use De Moivre's Theorem to express $\cos 4\theta$ in terms of $\cos \theta$ 2
- ii) Use the result above to solve the equation $8x^4 - 8x^2 + 1 = 0$ 2
- iii) Hence, show that $\cos \frac{\pi}{8} \cos \frac{3\pi}{8} \cos \frac{5\pi}{8} \cos \frac{7\pi}{8} = \frac{1}{8}$. 1

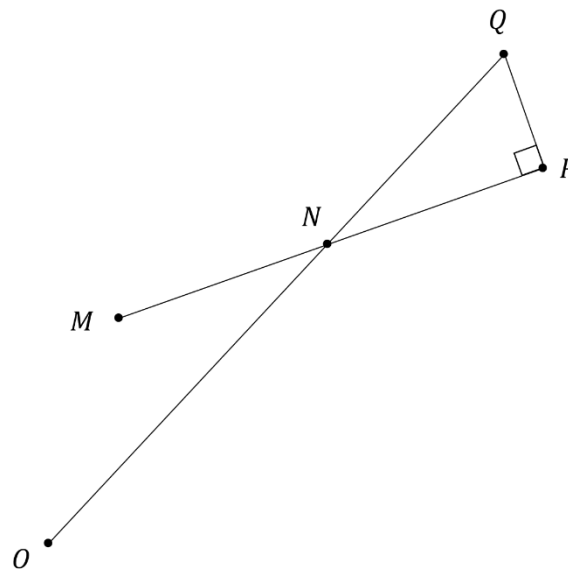
Question 16 continued on next page.

Question 16 continued.

- d. M and P are two points with position vectors $\overrightarrow{OM} = \underset{\sim}{m}$ and $\overrightarrow{OP} = \underset{\sim}{p}$ respectively. 3

N is the midpoint of MP and has position vector $\overrightarrow{ON} = \underset{\sim}{n}$.

PQ is perpendicular to MN and $\overrightarrow{NQ} = \lambda \overrightarrow{ON}$.



NOT TO
SCALE

Show that

$$\lambda = \frac{|\underset{\sim}{m}|^2 + |\underset{\sim}{n}|^2 - 2\underset{\sim}{m} \cdot \underset{\sim}{n}}{|\underset{\sim}{n}|^2 - \underset{\sim}{m} \cdot \underset{\sim}{n}}.$$

End of Question 16

End of Paper

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Student Number

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439

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2024

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

Writing Booklet

~~Question 11~~

Multiple Choice



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11

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Additional blank writing booklets set: 1 of 2

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Start here for
Question Number:

11

Ext 2 Trial 2025 Solns

Multiple choice :

1. By obs. A

A D A B C

D B D D A

2. $A \rightarrow B \begin{pmatrix} -6 \\ -3 \\ -6 \end{pmatrix} \Rightarrow D$

3. By obs. A

4. $v^2 = n^2(a^2 - x^2)$

$25 = n^2(9 - 4)$

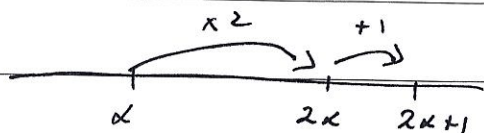
$n^2 = 5$

$T = \frac{2\pi}{n}$

$= \frac{2\pi}{\sqrt{5}}$

$= \frac{2\sqrt{5}\pi}{5} \Rightarrow B$

5.



To get these roots, $f(x)$ has been :

- horizontally dilated by sf of 2 $\Rightarrow f\left(\frac{x}{2}\right)$

- horizontally translated to $\Rightarrow f\left(\frac{x-1}{2}\right) \Rightarrow C$
right by 1

6. $z = x + iy$
 $z^2 = (x^2 - y^2) + 2xyi$
 $= a + bi$

$x < y$
 $\therefore x^2 > y^2$ since both neg.
 $x^2 - y^2 > 0$
 ie $a > 0$

$x < 0, y < 0$
 $\therefore xy > 0$
 ie $b > 0$

$\Rightarrow D$

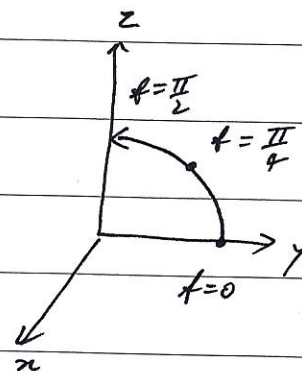
7. spiral about x -axis $\Rightarrow$ not A
 rate will increase $\Rightarrow$ not D

$t=0, y=2, z=0$

$t=\frac{\pi}{4}, y=\sqrt{2}, z=\sqrt{2}$

$t=\frac{\pi}{2}, y=0, z=2$

anticlockwise $\Rightarrow B$

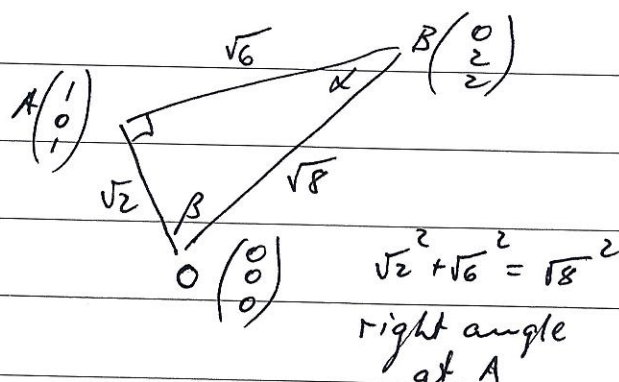


8.

$\tan \alpha = \frac{\sqrt{2}}{\sqrt{6}}$

$\alpha = \frac{\pi}{6} \therefore \beta = \frac{\pi}{3}$

ie $\angle AOB = 2\angle ABO \Rightarrow D$



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9. $r^3 \text{cis } 3\theta = -p^3 \text{cis}(\frac{\pi}{2})$

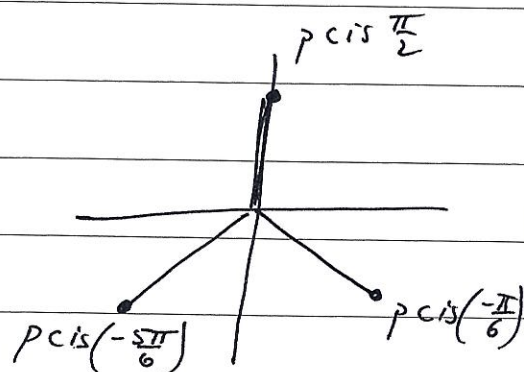
$= p^3 \text{cis}(\frac{3\pi}{2})$

$\therefore r=p, \quad 3\theta = \frac{3\pi}{2} + 2k\pi$

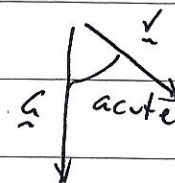
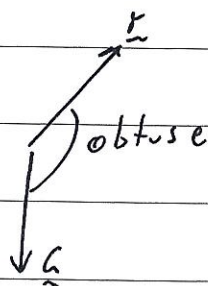
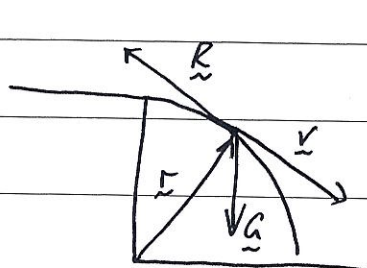
$k=0, \quad \theta = \frac{\pi}{2}$

Im parts $= 2 \times p \times \sin(-\frac{\pi}{6})$

$= -p \Rightarrow D$



10.



$\underline{u} \cdot \underline{v} = 0$

$\underline{R} \cdot \underline{u} = 0$

$\frac{\underline{R} \cdot \underline{v}}{|\underline{R}| |\underline{v}|} = -1$

$\frac{\underline{u} \cdot \underline{u}}{|\underline{u}| |\underline{u}|} < 0$ since obtuse

$\frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} > 0$ since acute

A is only option to fit all. $\Rightarrow A$



⇐ Tick this box if you have continued this answer in another writing booklet.

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Mathematics Extension 2

Writing Booklet

Question 11

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Start here for
Question Number:

11

$$a. \quad |z| = \sqrt{p^2 + (-4)^2 + 5^2} \quad (1)$$

$$5\sqrt{2} = \sqrt{p^2 + 16 + 25}$$

$$50 = p^2 + 41$$

$$p^2 = 9$$

$$p = \pm 3 \quad (1)$$

$$b. \quad \underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

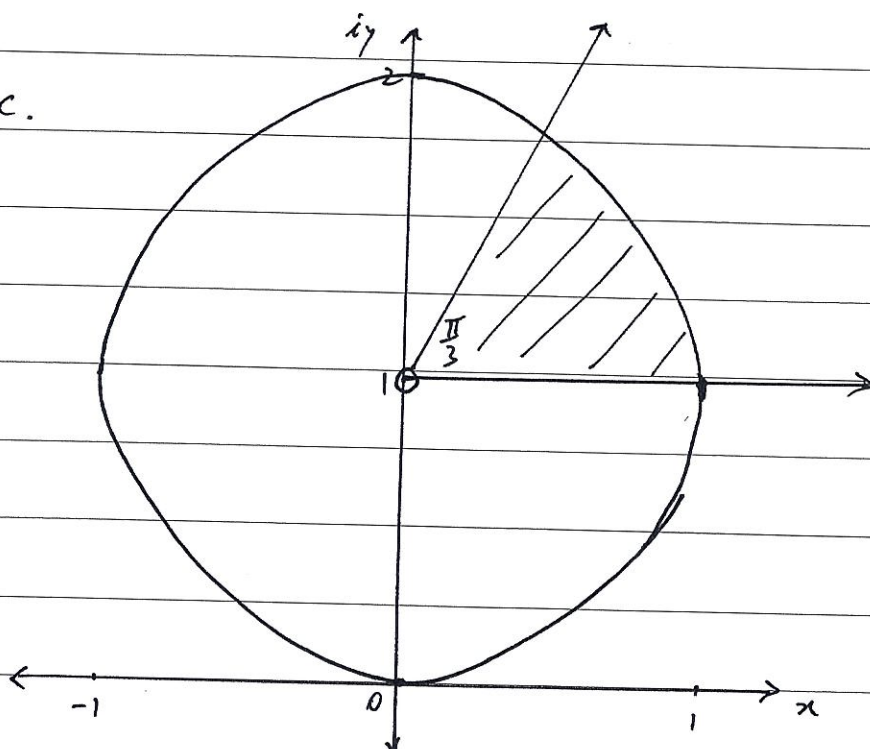
$$\cos \theta = \frac{1 \times 3 + (-2)(-2) + 2(-1)}{\sqrt{1^2 + (-2)^2 + 2^2} \times \sqrt{3^2 + (-2)^2 + (-1)^2}} \quad (1)$$

$$= \frac{5}{3 \times \sqrt{14}}$$

$$\theta = 63^\circ 32' 55.68''$$

$$= 63^\circ 33' \text{ (nearest min)} \quad (1)$$

c.



(1) mod

(1) arg

(1) open circle
solid lines
shading.

d. Consider $(a-b)^2 \geq 0$ ①

$$a^2 - 2ab + b^2 \geq 0$$

$$a^2 + b^2 \geq 2ab$$

$$\frac{a^2}{ab} + \frac{b^2}{ab} \geq 2$$

$$\frac{a}{b} + \frac{b}{a} \geq 2 \quad ①$$

or, by contradiction, or other methods.

e. $z^2 = 4 \operatorname{cis}\left(-\frac{2\pi}{3}\right)$

$$= 4e^{-\frac{2\pi}{3}i}$$

$$z = \pm \left(4e^{-\frac{2\pi}{3}i}\right)^{\frac{1}{2}} \quad ①$$

$$= \pm 2e^{-\frac{\pi}{3}i}$$

but only $z = 2e^{-\frac{\pi}{3}i}$ lies in the 4th quadrant.
 $= 2 \operatorname{cis}\left(-\frac{\pi}{3}\right) \quad ①$

f. If p is an integer, then it could be odd or even.

If p is odd, then both $(p+5)$ and $(p+3)$ are even

$\therefore (p+5)^{36}$ and $(p+3)^{35}$ are even ①
 and the sum of two even numbers cannot be odd

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$\therefore$ There is no odd p solution.

If p is even, then both $(p+5)$ and $(p+3)$ are odd
 $\therefore (p+5)^{36}$ and $(p+3)^{35}$ are both odd (1)
 and the sum of two odd numbers is even

$\therefore$ There is no integer solution

$\therefore$ There is no integer root for the eqn.

$$g. \int x \ln x \, dx = \int \frac{d\left(\frac{x^2}{2}\right)}{dx} \ln x \, dx \quad (1)$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \int x^2 \times \frac{1}{x} \, dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \int x \, dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \times \frac{x^2}{2} + c$$

$$= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + c \quad (1)$$



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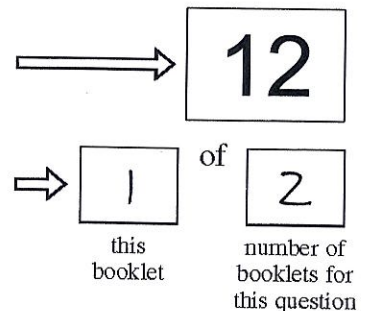
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Question 12



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$$a) \frac{x}{(x-2)^2(x-1)} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x-1}$$

$$x = A(x-2)(x-1) + B(x-1) + C(x-2)^2$$

$$x=1, \quad 1 = 0 + 0 + C$$

$$\therefore C = 1$$

$$x=2, \quad 2 = 0 + B + 0$$

$$\therefore B = 2$$

$$x=0, \quad 0 = A(-2)(-1) + 2(-1) + 1(-2)^2$$

$$0 = 2A - 2 + 4$$

$$2A = -2$$

$$A = -1$$

① some

① all

$$\therefore \frac{x}{(x-2)^2(x-1)} = \frac{-1}{x-2} + \frac{2}{(x-2)^2} + \frac{1}{x-1}$$

$$ii) \int_3^4 \frac{x}{(x-2)^2(x-1)} dx = \int_3^4 \frac{-1}{x-2} + \frac{2}{(x-2)^2} + \frac{1}{x-1} dx$$

$$= \int_3^4 \frac{-1}{x-2} + 2(x-2)^{-2} + \frac{1}{x-1} dx$$

$$= \left[-\ln|x-2| - 2(x-2)^{-1} + \ln|x-1| \right]_3^4 \quad \text{①}$$

$$= -\ln|4-2| - \frac{2}{4-2} + \ln|4-1| - \left(-\ln|3-2| - \frac{2}{3-2} + \ln|3-1| \right)$$

$$= -\ln 2 - 1 + \ln 3 + \ln 1 + 2 - \ln 2 \quad \text{①}$$

$$= \ln 3 - 2 \ln 2 + 1$$

$$= \ln 3 - \ln 4 + 1$$

$$= \ln\left(\frac{3}{4}\right) + 1 \quad (1)$$

b. Converse is: if n is odd, then $2n^2 + 3n + 1$ is even.

Let $n = 2k+1$ where $k \in \mathbb{Z}$ (1)

$$\therefore 2n^2 + 3n + 1 = 2(2k+1)^2 + 3(2k+1) + 1$$

$$= 2(4k^2 + 4k + 1) + 6k + 3 + 1$$

$$= 8k^2 + 8k + 2 + 6k + 4$$

$$= 8k^2 + 14k + 6$$

$$= 2(4k^2 + 7k + 3) \quad (1)$$

$= 2P$ where $P = 4k^2 + 7k + 3$ and must be an integer since $k \in \mathbb{Z}$

which is even

$\therefore$ The converse is true.

c. Since $P(z)$ is real, $2+3i$ must also be a zero (1)

Let the zeroes be $\alpha, \beta, 2-3i, 2+3i$

$$\therefore \alpha + \beta + 2 - 3i + 2 + 3i = 6$$

$$\alpha + \beta = 2$$

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$$\therefore \beta = 2 - \alpha$$

$$\alpha\beta(2+3i)(2-3i) = -39$$

$$\alpha\beta(4+9) = -39$$

$$\alpha\beta = -3$$

(1)

$$\alpha(2-\alpha) = -3$$

$$2\alpha - \alpha^2 = -3$$

$$\alpha^2 - 2\alpha - 3 = 0$$

$$(\alpha+1)(\alpha-3) = 0$$

$$\alpha = -1 \quad \text{or} \quad \alpha = 3$$

$$\text{ie } \beta = 3 \quad \beta = -1$$

$$\therefore P(z) = (z+1)(z-3)(z-(2-3i))(z-(2+3i))$$

$$= (z+1)(z-3)(z-2+3i)(z-2-3i)$$

(1)

$$\text{d. } \phi = \frac{2re^{i\theta}}{(re^{i\theta})^2 + r^2}$$

$$= \frac{2re^{i\theta}}{r^2(e^{2i\theta} + 1)}$$

(1)

$$= \frac{2e^{i\theta}}{re^{i\theta}(e^{i\theta} + e^{-i\theta})}$$

$$= \frac{2}{r(z + \bar{z})}$$



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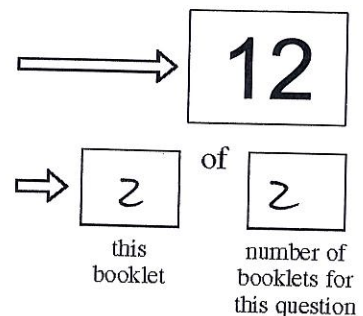
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$$d. (cont) \quad = \frac{2}{1 \times 2 \operatorname{Re}(z)} \quad (1)$$

which must be real.

$$\begin{aligned} e. \quad \int \frac{e^{2x} - e^x}{e^x + 1} dx &= \int \frac{e^x (e^x - 1)}{e^x + 1} dx \\ &= \int \frac{e^x (e^x + 1 - 2)}{e^x + 1} dx \quad (1) \\ &= \int \frac{e^x (e^x + 1)}{e^x + 1} - \frac{2e^x}{e^x + 1} dx \\ &= \int e^x dx - 2 \int \frac{e^x}{e^x + 1} dx \quad (1) \\ &= e^x - 2 \ln |e^x + 1| + c \\ &= e^x - 2 \ln(e^x + 1) + c \quad (1) \end{aligned}$$

since $e^x > 0$

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Question 13



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$$\begin{aligned} \text{a. i)} \quad & (a-b)^2 > 0 \\ & a^2 - 2ab + b^2 > 0 \\ & a^2 - ab + b^2 > ab \end{aligned}$$

$$\begin{aligned} \text{Now } a^3 + b^3 &= (a+b)(a^2 - ab + b^2) \\ &> (a+b)ab \\ &> ab(a+b) \end{aligned} \quad \textcircled{1}$$

$$\begin{aligned} \text{ii)} \quad & \text{Similarly,} \\ & b^3 + c^3 > bc(b+c) \\ & c^3 + a^3 > ca(c+a) \end{aligned}$$

$$\begin{aligned} \therefore 2(a^3 + b^3 + c^3) &> ab(a+b) + bc(b+c) + ca(c+a) \quad \textcircled{1} \\ &> a^2b + ab^2 + b^2c + bc^2 + ac^2 + a^2c \\ &> a^2b + a^2c + b^2a + b^2c + c^2a + c^2b \\ &> a^2(b+c) + b^2(a+c) + c^2(a+b) \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{b. i)} \quad x &= 5 - 4\cos^2 2\pi t \\ &= 5 - 4 \left(\frac{1 + \cos 4\pi t}{2} \right) \quad \textcircled{1} \\ &= 5 - 2(1 + \cos 4\pi t) \\ &= 3 - 2\cos 4\pi t \end{aligned}$$

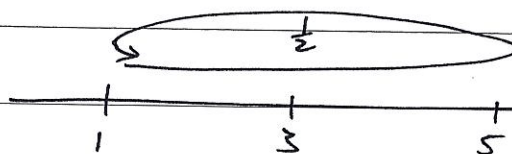
$$\begin{aligned} \therefore \text{centre of motion: } x &= 3 \quad \textcircled{1} \\ \text{amplitude is } &2 \quad \textcircled{1} \end{aligned}$$

$$11) \quad T = \frac{2\pi}{a}$$

$$= \frac{2\pi}{4\pi}$$

$$= \frac{1}{2} \text{ sec}$$

$\therefore$ in 5 seconds, will complete 10 complete oscillations ^①



one oscillation is 8cm

$\therefore$ travels 80 cm in first 5 seconds. ^①

c. Assume that $P(x)$ has a double root at $x = \alpha$

$$\therefore P(\alpha) = P'(\alpha) = 0 \quad \text{①}$$

$$\text{or } P(\alpha) - P'(\alpha) = 0$$

$$\text{ie. } \alpha^5 + 5\alpha^4 + 20\alpha^3 + 60\alpha^2 + 120\alpha + 120 - (5\alpha^4 + 20\alpha^3 + 60\alpha^2 + 120\alpha + 120) = 0$$

$$\text{ie } \alpha^5 = 0$$

$$\alpha = 0 \quad \text{①}$$

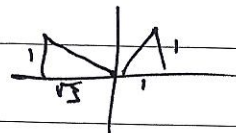
but, if $\alpha = 0$ then $P(\alpha) = 120$ ^①

and, hence, $\alpha = 0$ cannot be a root

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which is a contradiction
 $\therefore P(x)$ has no double roots.

d. $z = 1+i$ $w = -\sqrt{3} + i$



i) $zw = \sqrt{2} \operatorname{cis} \frac{\pi}{4} \times 2 \operatorname{cis} \frac{5\pi}{6}$ ①

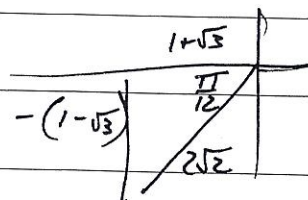
$$= 2\sqrt{2} \operatorname{cis} \left(\frac{13\pi}{12} \right)$$

$$= 2\sqrt{2} \operatorname{cis} \left(-\frac{11\pi}{12} \right)$$
 ①

ii) $zw = (1+i)(-\sqrt{3}+i)$

$$= (-\sqrt{3}-1) - \sqrt{3}i + i$$

$$= -(1+\sqrt{3}) + i(1-\sqrt{3})$$
 ①



$$\therefore \cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$
 ①



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$$\begin{aligned}
 a. \quad \int \frac{2x-5}{x^2+2x+2} dx &= \int \frac{2x+2-7}{x^2+2x+2} dx \quad (1) \\
 &= \int \frac{2x+2}{x^2+2x+2} dx - 7 \int \frac{1}{(x^2+2x+1)+1} dx \quad (1) \\
 &= \ln|x^2+2x+2| - 7 \int \frac{1}{1+(x+1)^2} dx \\
 &= \ln|x^2+2x+2| - 7 \tan^{-1}(x+1) + C \quad (1)
 \end{aligned}$$

$$b. \quad 1) \quad \ddot{x} = \frac{1000-v}{800}$$

$$\frac{dv}{dt} = \frac{1000-v}{800}$$

$$\frac{800}{1000-v} dv = dt$$

$$-800 \int_0^v \frac{-1}{1000-v} dv = \int_0^t dt \quad (1)$$

$$-800 \left[\ln|1000-v| \right]_0^v = t \quad (1)$$

$$t = -800 \left[\ln|1000-v| - \ln 1000 \right]$$

$$= 800 \left(\ln 1000 - \ln(1000-v) \right) \quad (1)$$

since $v < 1000$

$$= 800 \ln \left(\frac{1000}{1000-v} \right)$$

11) when $t = 10$

$$10 = 800 \ln \left(\frac{1000}{1000 - v} \right) \quad (1)$$

$$e^{\frac{1}{80}} = \frac{1000}{1000 - v} \quad (1)$$

$$1000 - v = 1000 e^{-\frac{1}{80}}$$

$$v = 1000 - 1000 e^{-\frac{1}{80}}$$

$$= 12.42219951$$

$$= 12.4 \text{ m/s (1dp)} \quad (1)$$

c. $\underline{u} = \underline{a} + m\underline{b} \quad \underline{v} = m\underline{a} - \underline{b}$

$$|\underline{a}| = |\underline{b}| = 1 \quad \underline{u} \cdot \underline{v} = 1$$

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \frac{\pi}{3}$$

$$= 1 \times 1 \times \frac{1}{2}$$

$$= \frac{1}{2} \quad (1)$$

$$(\underline{a} + m\underline{b}) \cdot (m\underline{a} - \underline{b}) = 1 \quad (1)$$

$$m\underline{a} \cdot \underline{a} - \underline{a} \cdot \underline{b} + m^2 \underline{a} \cdot \underline{b} - m\underline{b} \cdot \underline{b} = 1$$

$$m|\underline{a}|^2 + (m^2 - 1)\underline{a} \cdot \underline{b} - m|\underline{b}|^2 = 1$$

$$m \times 1 + (m^2 - 1) \times \frac{1}{2} - m \times 1 = 1$$

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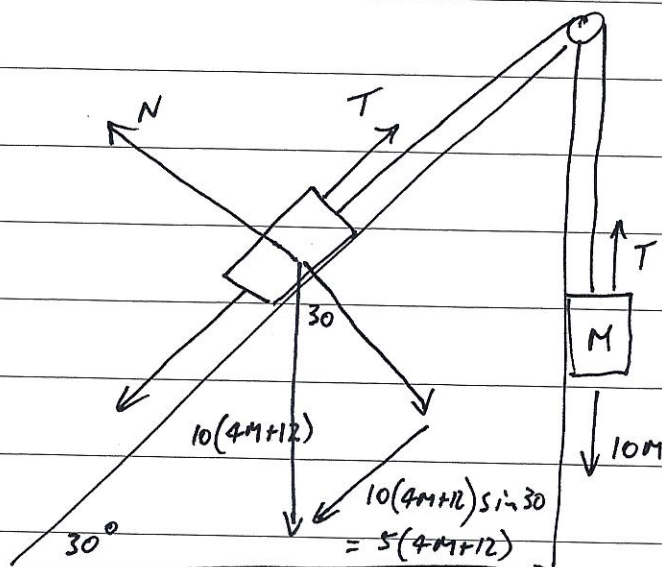
$$m^2 - 1 = 2$$

$$m^2 = 3$$

$$m = \pm\sqrt{3}$$

①

d.



For the system:

$$\sum F = \sum m\ddot{x}$$

$$\text{ie } (5M+12)3 = 5(4M+12) - T + T - 10M \quad (1)$$

$$15M + 36 = 20M + 60 - 10M$$

$$15M + 36 = 10M + 60$$

$$5M = 24$$

$$M = \frac{24}{5}$$

$$= 4.8 \text{ kg} \quad (1)$$



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a. 1) $f(t) = \frac{t \sin t}{2}$ (1)

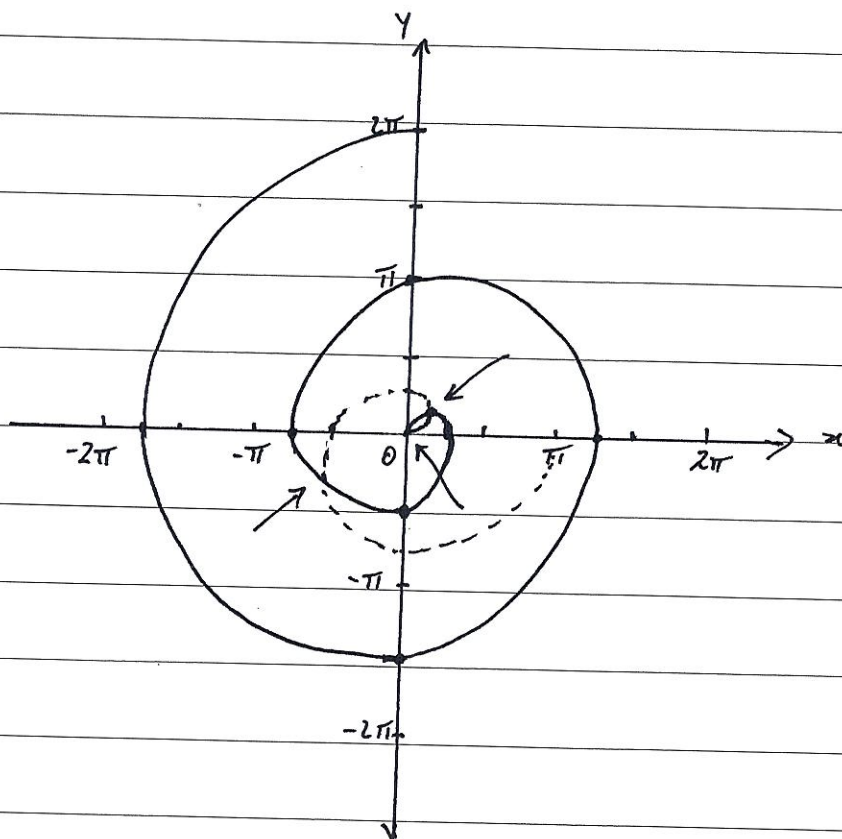
$g(t) = \frac{t \cos t}{2}$ (1)

dist. from 0 is always $\frac{t}{2}$

$\therefore T = 4\pi$ (1)

ii) $x = g(t) = \frac{t \cos t}{2}$
 $y = f(t) = \frac{t \sin t}{2}$ $[0, 2\pi]$

t	x	y
0	0	0
$\frac{\pi}{4}$	$\frac{\pi}{8} \times \frac{\sqrt{2}}{2}$ (+ve)	$\frac{\pi}{8} \times \frac{\sqrt{2}}{2}$ (+ve)
$\frac{\pi}{2}$	0	$\frac{\pi}{4}$
π	$-\frac{\pi}{2}$	0
$\frac{3\pi}{2}$	0	$-\frac{3\pi}{4}$
2π	π	0



There are 3 pts of intersection. (1)

b. 1) $I_n = \int_0^1 \sqrt{x} (1-x)^n dx$

$$= \int_0^1 \frac{d\left(\frac{2x^{\frac{3}{2}}}{3}\right)}{dx} (1-x)^n dx \quad (1)$$

$$= \left[\frac{2x^{\frac{3}{2}}}{3} (1-x)^n \right]_0^1 - \frac{2}{3} \int_0^1 x^{\frac{3}{2}} \times n(1-x)^{n-1} x^{-1} dx$$

$$= 0 - \frac{2n}{3} \int_0^1 -x \sqrt{x} (1-x)^{n-1} dx$$

$$= -\frac{2n}{3} \int_0^1 (1-x-1) \sqrt{x} (1-x)^{n-1} dx \quad (1)$$

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$$= -\frac{2n}{3} \int_0^1 (1-x) \sqrt{x} (1-x)^{n-1} - 1 \sqrt{x} (1-x)^{n-1} dx$$

$$= -\frac{2n}{3} \int_0^1 \sqrt{x} (1-x)^n - \sqrt{x} (1-x)^{n-1} dx$$

$$\text{ie } I_n = -\frac{2n}{3} I_n + \frac{2n}{3} I_{n-1}$$

$$I_n \left(1 + \frac{2n}{3} \right) = \frac{2n}{3} I_{n-1}$$

$$I_n \left(\frac{3+2n}{3} \right) = \frac{2n}{3} I_{n-1} \quad (1)$$

$$I_n = \frac{2n}{2n+3} I_{n-1}$$

$$\text{ii) } I_1 = \frac{2}{2+3} I_0$$

$$= \frac{2 I_0}{5}$$

$$I_0 = \int_0^1 \sqrt{x} dx$$

$$= \left[\frac{2x^{\frac{3}{2}}}{3} \right]_0^1$$

$$= \frac{2}{3}$$



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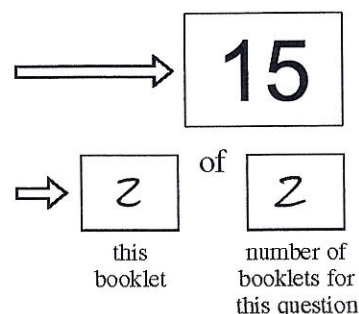
Question 15

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b. ii) (cont) $\therefore I_1 = \frac{2}{5} \times \frac{2}{3}$

$$= \frac{4}{15}$$

(1)

c. i) $\sum F = \sum m\ddot{x}$

$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2 \quad (1)$$

ii) achieves V_T when $\ddot{x} = 0$

$$\text{ie } g = kv^2$$

$$v^2 = \frac{g}{k}$$

$$v = \sqrt{\frac{g}{k}} \quad \text{since } v > 0 \quad (1)$$

iii) $v \frac{dv}{dx} = g - kv^2$

$$\frac{v}{g - kv^2} dv = dx$$

$$-\frac{1}{2k} \times \frac{-2kv}{g - kv^2} dv = dx$$

$$-\frac{1}{2k} \int_0^v \frac{-2kv}{g - kv^2} dv = \int_0^x dx \quad (1)$$

$$-\frac{1}{2k} \left[\ln |g - kv^2| \right]_0^v = x \quad (1)$$

$$\begin{aligned}
 x &= \frac{-1}{2k} \left[\ln|g - kv^2| - \ln g \right] \\
 &= \frac{1}{2k} \left[\ln g - \ln|g - kv^2| \right] \quad (1)
 \end{aligned}$$

and, since $g - kv^2 > 0$

$$x = \frac{1}{2k} \ln \left(\frac{g}{g - kv^2} \right)$$

iv) when $v = \frac{1}{2} \sqrt{\frac{g}{k}}$

$$v^2 = \frac{g}{4k} \quad (1)$$

$$\begin{aligned}
 \therefore x &= \frac{1}{2k} \ln \left(\frac{g}{g - k \times \frac{g}{4k}} \right) \\
 &= \frac{1}{2k} \ln \left(\frac{g}{\frac{3g}{4}} \right) \\
 &= \frac{1}{2k} \ln \left(\frac{4}{3} \right) \quad (1)
 \end{aligned}$$

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$$a. \quad \vec{r} = \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$\therefore x = -1 + \lambda, \quad y = 2 + 2\lambda, \quad z = 6 + \lambda$$

At pts of intersection:

$$(-1 + \lambda + 1)^2 + (2 + 2\lambda - 2)^2 + (6 + \lambda - 3)^2 = 45 \quad (1)$$

$$\lambda^2 + 4\lambda^2 + (3 + \lambda)^2 = 45$$

$$5\lambda^2 + 9 + 6\lambda + \lambda^2 = 45$$

$$6\lambda^2 + 6\lambda - 36 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

$$(\lambda - 2)(\lambda + 3) = 0$$

$$\lambda = 2 \quad \text{or} \quad \lambda = -3$$

 $\therefore$ Pts of int at:

$$A \begin{pmatrix} 1 \\ 6 \\ 8 \end{pmatrix} \quad \text{and} \quad B \begin{pmatrix} -4 \\ -4 \\ 3 \end{pmatrix} \quad (1)$$

Shortest distance from C will be to M,
midpoint of AB

$$M \text{ is } \begin{pmatrix} -\frac{3}{2} \\ 1 \\ \frac{11}{2} \end{pmatrix}$$

$$C \text{ is } \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} \quad (1)$$

$$MC = \sqrt{\left(-\frac{3}{2} + 1\right)^2 + (1 - 2)^2 + \left(\frac{11}{2} - 3\right)^2}$$

$$= \sqrt{\frac{1}{4} + 1 + \frac{25}{4}}$$

$$= \sqrt{\frac{30}{4}}$$

$$= \frac{\sqrt{30}}{2}$$

①

b. $a_{n+1} = \sqrt{2a_n}$ $a_1 = 1$

RTP: $a_n = 2^{(1-2^{(1-n)})}$ for $n \geq 1$

Prove true for $n=1$

$$a_1 = 2^{(1-2^{(1-1)})}$$

$$= 2^{1-1}$$

$$= 2^0$$

$$= 1$$

①

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\text{ie } a_k = 2^{(1-2^{(1-k)})}$$

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Prove true for $n = k+1$

$$\text{ie } a_{k+1} = 2^{(1-2^{(1-(k+1))})}$$

$$= 2^{(1-2^{-k})}$$

Now, $a_{k+1} = \sqrt{2 a_k}$

$$= \sqrt{2 \times 2^{(1-2^{(1-k)})}} \quad (1) \quad \text{from assumption}$$

$$= \sqrt{2^{(1-2^{(1-k)})+1}}$$

$$= \sqrt{2^{(2-2^{(1-k)})}}$$

$$= \sqrt{2^{(2'-2' \times 2^{-k})}}$$

$$= \sqrt{\frac{2(1-2^{-k})}{2}}$$

$$= \left(\frac{2(1-2^{-k})}{2} \right)^{\frac{1}{2}}$$

$$= 2^{1-2^{-k}} \quad (1)$$



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EXAMINATION**

Mathematics Extension 2

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Question 16

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16 b. (cont)

= RHS

 $\therefore$ true for $n = k + 1$ $\therefore$ by induction, the statement is true for all integers $n \geq 1$

$$c. \quad i) \quad \text{cis } 4\theta = (\cos \theta + i \sin \theta)^4$$

$$= \cos^4 \theta + 4 \cos^3 \theta i \sin \theta + 6 \cos^2 \theta i^2 \sin^2 \theta$$

$$+ 4 \cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta \quad (1)$$

Equating reals:

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta$$

$$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1 \quad (1)$$

$$ii) \quad 8x^4 - 8x^2 + 1 = 0$$

$$\text{let } x = \cos \theta$$

$$\therefore 8 \cos^4 \theta - 8 \cos^2 \theta + 1 = 0$$

$$\text{ie } \cos 4\theta = 0$$

$$\therefore 4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \dots \quad (1)$$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \dots$$

$\therefore$ The 4 distinct solutions are :

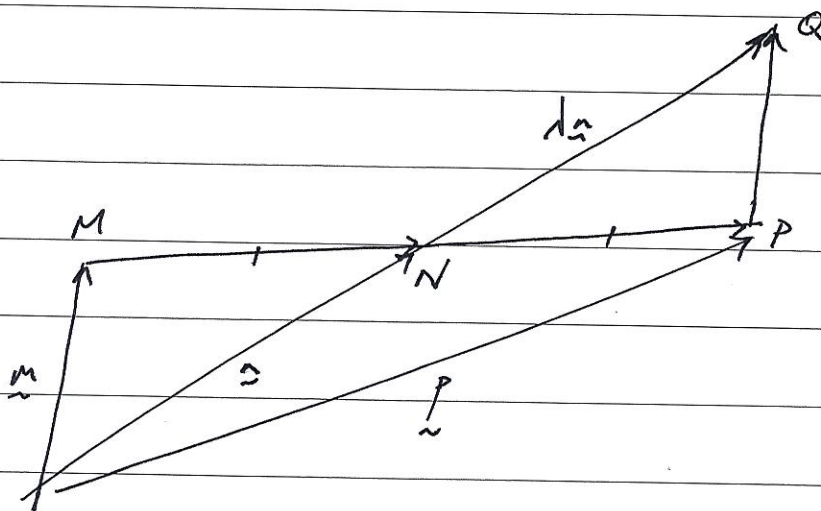
$$x = \cos \frac{\pi}{8}, \quad x = \cos \frac{3\pi}{8}, \quad x = \cos \frac{5\pi}{8}, \quad x = \cos \frac{7\pi}{8} \quad (1)$$

iii) Product of roots : $\frac{c}{a} = \frac{1}{8}$

(1)

$$\text{i.e. } \cos \frac{\pi}{8} \cos \frac{3\pi}{8} \cos \frac{5\pi}{8} \cos \frac{7\pi}{8} = \frac{1}{8}$$

d.



$$\vec{OM} + \vec{MP} = \vec{OP}$$

$$\vec{MP} = \vec{OP} - \vec{OM}$$

$$= \underline{p} - \underline{m}$$

(1) some prep

$$\vec{OP} + \vec{PQ} = \vec{OQ}$$

$$\vec{PQ} = \vec{OQ} - \vec{OP}$$

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$$\vec{PQ} = (1+\lambda)\vec{n} - \vec{p}$$

Now, $\vec{MP} \perp \vec{PQ}$

$$\therefore \vec{MP} \cdot \vec{PQ} = 0$$

$$\text{ie } (\vec{p} - \vec{m}) \cdot ((1+\lambda)\vec{n} - \vec{p}) = 0 \quad (1)$$

$$(1+\lambda)\vec{p} \cdot \vec{n} - (1+\lambda)\vec{m} \cdot \vec{n} + \vec{p} \cdot \vec{m} - \vec{p} \cdot \vec{p} = 0$$

$$(1+\lambda)[\vec{p} \cdot \vec{n} - \vec{m} \cdot \vec{n}] = \vec{p} \cdot \vec{p} - \vec{p} \cdot \vec{m}$$

$$(1+\lambda) = \frac{\vec{p} \cdot \vec{p} - \vec{p} \cdot \vec{m}}{\vec{p} \cdot \vec{n} - \vec{m} \cdot \vec{n}}$$

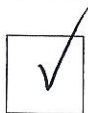
Now, $\vec{OM} + \vec{MN} = \vec{ON}$

$$\vec{m} + \frac{1}{2}(\vec{p} - \vec{m}) = \vec{n}$$

$$\frac{1}{2}\vec{p} + \frac{1}{2}\vec{m} = \vec{n}$$

$$\vec{p} + \vec{m} = 2\vec{n}$$

$$\vec{p} = 2\vec{n} - \vec{m}$$



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d. (cont)

$$\therefore 1+\lambda = \frac{(2\underline{n}-\underline{m}) \cdot (2\underline{n}-\underline{m}) - (2\underline{n}-\underline{m}) \cdot \underline{m}}{(2\underline{n}-\underline{m}) \cdot \underline{n} - \underline{m} \cdot \underline{n}}$$

$$= \frac{4\underline{n} \cdot \underline{n} - 4\underline{m} \cdot \underline{n} + \underline{m} \cdot \underline{m} - 2\underline{m} \cdot \underline{n} + \underline{m} \cdot \underline{m}}{2\underline{n} \cdot \underline{n} - 2\underline{m} \cdot \underline{n}}$$

$$= \frac{4|\underline{n}|^2 - 6\underline{m} \cdot \underline{n} + 2|\underline{m}|^2}{2|\underline{n}|^2 - 2\underline{m} \cdot \underline{n}}$$

$$1+\lambda = \frac{2|\underline{n}|^2 + |\underline{m}|^2 - 3\underline{m} \cdot \underline{n}}{|\underline{n}|^2 - \underline{m} \cdot \underline{n}}$$

$$\therefore \lambda = \frac{2|\underline{n}|^2 + |\underline{m}|^2 - 3\underline{m} \cdot \underline{n} - |\underline{n}|^2 + \underline{m} \cdot \underline{n}}{|\underline{n}|^2 - \underline{m} \cdot \underline{n}}$$

$$= \frac{|\underline{n}|^2 + |\underline{m}|^2 - 2\underline{m} \cdot \underline{n}}{|\underline{n}|^2 - \underline{m} \cdot \underline{n}} \quad (1)$$